In [28]:
```
#Import statements
import pandas as pd
import numpy as np
import mlba
import seaborn as sns
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt
%matplotlib inline
```

In [2]:
```
#loading dataset
mower_df = mlba.load_data('RidingMowers.csv')
mower_df.head()
```

Out[2]:

	Income	Lot_Size	Ownership
0	60.0	18.4	Owner
1	85.5	16.8	Owner
2	64.8	21.6	Owner
3	61.5	20.8	Owner
4	87.0	23.6	Owner

a. What percentage of households in the study were owners of a riding mower?

In [3]:
```
print(mower_df.columns)
```

```
Index(['Income', 'Lot_Size', 'Ownership'], dtype='object')
```

In [21]:
```
pct_owners = mower_df["Ownership"].mean() * 100
print(f"Percentage of owners: {pct_owners:.2f}%")
```

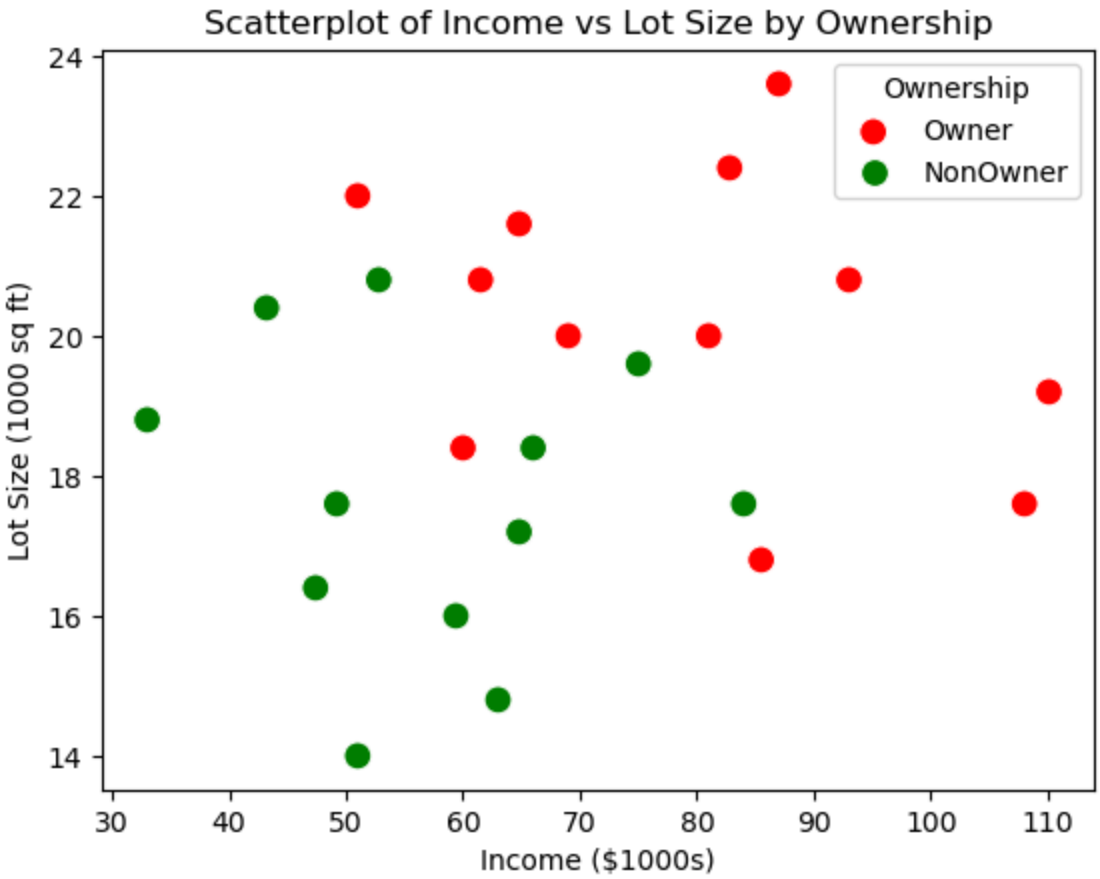
```
Percentage of owners: 50.00%
```

b. Create a scatter plot of Income vs. Lot Size using color or symbol to distinguish owners from nonowners. From the scatter plot, which class seems to have a higher average income, owners or nonowners?

In [4]:
```
mower_df["Ownership"] = mower_df["Ownership"].map({"Owner": 1, "Nonowner": 0
```

In [5]:
```
# Owners (red) generally have higher income and larger lots than Nonowners (
sns.scatterplot(x="Income", y="Lot_Size", hue="Ownership",
                data=mower_df, palette={0:"green", 1:"red"}, s=100)
# Scatter plot showing Income vs Lot Size.
plt.title("Scatterplot of Income vs Lot Size by Ownership")
plt.xlabel("Income ($1000s)")
plt.ylabel("Lot Size (1000 sq ft)")
```

```
plt.legend(title="Ownership", labels=["Owner","NonOwner"])
plt.show()
```



In [6]:
```
# Defining predictors (Income, Lot Size) as X and target variable (Ownership
X = mower_df[["Income", "Lot_Size"]]
y = mower_df["Ownership"]
```

In [7]:
```
log_reg = LogisticRegression()
log_reg.fit(X, y)
```

Out[7]:
```
▾ LogisticRegression
LogisticRegression()
```

In [8]:
```
# Print logistic regression intercept and coefficients
print("Intercept:", log_reg.intercept_)
print("Coefficients:", log_reg.coef_)
```

```
Intercept: [-22.32789857]
Coefficients: [[0.10085067 0.81169613]]
```

c. Among nonowners, what is the percentage of households classified correctly?

In [9]:
```
# Calculate the percentage of Nonowners correctly classified by the model.
y_pred = log_reg.predict(X)
```

```
nonowners = mower_df[y == 0]
nonowner_correct = np.mean(y_pred[y == 0] == 0) * 100
print(f"Percentage of correctly classified Nonowners: {nonowner_correct:.1f}
```

```
Percentage of correctly classified Nonowners: 83.3%
```

d. To increase the percentage of correctly classified nonowners, should the threshold probability be increased or decreased?

In [10]:
```
# Effect of threshold adjustment
probs = log_reg.predict_proba(X)[:,1]
# To adjust, use predict_proba and set your own cutoff
y_pred_06 = (probs > 0.5).astype(int)
nonowner_correct_06 = np.mean(y_pred_06[y == 0] == 0) * 100
print(f"Correctly classified Nonowners (cutoff=0.6): {nonowner_correct_06:.1
```

```
Correctly classified Nonowners (cutoff=0.6): 83.3%
```

Note: Increasing the cutoff increases Nonowner accuracy.

e. What are the odds that a household with a $60K income and a lot size of 20,000 ftMath equation is an owner?

f. What is the classification of a household with a $60K income and a lot size of 20,000 ft^2 ? Use threshold = 0.5.

In [25]:
```
# Predict ownership for a household with Income=60 and Lot Size=20
X_new = pd.DataFrame({'Income': [60], 'Lot_Size': [20]})
# Compute logit, odds, probability
logit_val = log_reg.decision_function(X_new)[0]
odds_val = np.exp(logit_val)
prob_val = log_reg.predict_proba(X_new)[0,1]

# classification using threshold 0.5
classification = "Owner" if prob_val > 0.5 else "Nonowner"

print(f"Logit value: {logit_val:.3f}")
print(f"Odds of being Owner: {odds_val:.3f}")
print(f"Probability of being Owner: {prob_val:.3f}")
print(f"Classification (cutoff=0.5): {classification}")
```

```
Logit value: -0.043
Odds of being Owner: 0.958
Probability of being Owner: 0.489
Classification (cutoff=0.5): Nonowner
```

g. What is the minimum income that a household with 16,000 ftMath equation lot size should have before it is classified as an

owner?

In [26]:
```
b0 = log_reg.intercept_[0]
b1, b2 = log_reg.coef_[0]

# Solve β0 + β1*Income + β2*16 = 0
income_cutoff = (-b0 - b2*16)/b1
print(f"Minimum income for Lot Size=16 to classify as Owner: {income_cutoff:
```

```
Minimum income for Lot Size=16 to classify as Owner: 92.62 ($1000s)
```

In [ ]: